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RESEARCH ARTICLE

AI-Powered MetaHuman Interviewer: Serious Game for Student Job Interview Skills

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee of King Mongkut' s Institute of Technology Ladkrabang under Application No. EC-KMITL_67_030.

ABSTRACT Effective interview preparation is critical for university graduates entering a competitive job market, yet traditional methods often lack sufficient realism and interactivity. This paper presents the development and evaluation of an innovative serious game, the Interactive Interview Assistant Application (IIAA), designed to address this gap. The IIAA simulates job interviews using an AI-driven digital human interviewer powered by MetaHuman technology, a Large Language Model (LLM) for dynamic dialogue, and integrated Speech-to-Text, Text-to-Speech, and Speech-to-Animation systems, offering a personalized and immersive practice environment. The system was evaluated with senior undergraduate students across three distinct career paths (UX/UI Design, Software Engineering, Game Development). The evaluation focused on system performance, overall usability, and qualitative user feedback. Results indicate that the IIAA successfully creates an engaging and realistic simulation, with users reporting high satisfaction regarding its ability to emulate real-world interview scenarios and provide valuable, actionable feedback. While identified performance latencies related to high-definition 3D animation and voice processing require further optimization, the study underscores the significant potential of this approach. This research demonstrates that sophisticated game-based tools can substantially enhance interview preparedness, thereby improving student confidence and readiness for professional employment.

INDEX TERMS Serious games, game-based learning (GBL), educational technology, simulation and gaming, instructional design.

I. INTRODUCTION

Navigating the increasingly competitive contemporary job market presents a significant challenge for newly graduating students, for whom effective interview preparation is paramount to successfully transitioning into the workforce. Traditional preparation methods, however, may not always provide sufficiently engaging, personalized, or realistic practice opportunities. Addressing this need, this paper introduces the Interactive Interview Assistant Application (IIAA), an advanced serious game designed to enhance crucial interview competencies through dynamic, simulated scenarios.

The IIAA leverages cutting-edge technologies to create an immersive training environment where users engage in mock interviews with a highly realistic digital human interviewer, rendered using Epic Games' MetaHuman [1] technology. This system is underpinned by a sophisticated backend incorporating a Large Language Model (LLM) that enables the dynamic generation of interview questions tailored to specific career paths—UI/UX Designer, Software Engineer, and Game Developer—thereby offering a personalized and contextually relevant practice experience. Key functionalities include real-time speech-to-text conversion, AI-driven response evaluation with constructive feedback, and synchronized speech-to-animation for the digital avatar. The primary aim of the IIAA is to provide graduating students, particularly within the Thai context, with an accessible, engaging, and controlled platform to refine their interviewing techniques, build confidence, and improve their overall preparedness for real-world job interviews.

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This paper details the design, development, and initial evaluation of the IIAA. Section II (Literature Review) provides a review of relevant literature concerning serious games, interview simulation technologies, and AI in education. Section III (Methodology) outlines the architectural design of the IIAA, the specific technologies employed in its development, and the methods used for its user-based evaluation. Section IV (Results) presents the empirical findings from this evaluation, encompassing system performance metrics and user feedback on usability and effectiveness. Section V (Discussion) interprets these results, considers their implications, acknowledges study limitations, and proposes directions for future research. Finally, Section VI (Conclusion) summarizes the key contributions of this work and reiterates the potential of the IIAA in enhancing interview readiness.

II. LITERATURE REVIEW

This literature review establishes the context and rationale for the IIAA. It examines current technologies for job interview training, serious games in professional skill development, digital humans in educational simulations, LLMs for enhanced interaction, and speech recognition in learning environments. The review concludes by synthesizing these areas to identify existing gaps and highlight the proposed system's novelty.

A. TECHNOLOGICAL ADVANCEMENTS IN JOB INTERVIEW TRAINING

Technological approaches to job interview training increasingly use artificial intelligence (AI) and machine learning (ML) to improve candidate preparation and performance. These systems provide personalized feedback and realistic simulations, fostering better communication skills and confidence.

Key AI-driven trends include prominent AI-powered feedback systems. Some research explores technologies like Generative Adversarial Networks (GANs) for identifying behavioral weaknesses and generating tailored feedback to enhance performance. Other platforms use real-time response analysis to dynamically adjust question difficulty and provide immediate feedback, refining interview competencies.

Comprehensive evaluation tools are another significant development. Khallelullah et al. [2] note that integrating video analysis with audio-to-text technologies enables detailed assessment of verbal and non-verbal communication, offering objective performance metrics. Mock interview simulators also increasingly create realistic practice environments; for example, Jadhav et al. [3] describe simulators with AI-driven interviewers and posture detection to enhance realism.

The development of intelligent systems for generating tailored interview questions represents a significant thrust in customizable training applications. For example, DuerQues, a system by Qin et al. [4], employs a neural generative model and skill entity recognition to automatically create skill-oriented interview questions, recommending suitable queries based on specified skills. Similarly, a job interview simulation system designed for the Hong Kong professional environment by Lee et al. [5] incorporates AI-driven question generation to help graduates practice in multiple languages. Further advancing this area, Mishra et al. [6] describe an AI-driven virtual mock interview platform utilizing GPT-4 for automated mock interviews, which provides personalized feedback and generates context-aware questions. Another approach, EZInterviewer by Li et al. [7], uses AI to generate questions by analyzing job descriptions and candidate resumes, employing knowledge selectors and dialog generators to create relevant and coherent interview content. While these AI-driven tools significantly enhance the customization and relevance of interview practice, the challenge of fully replicating the nuanced dynamics and subtle interpersonal cues of human interaction in real-world interview settings remains an important consideration for ongoing research.

B. THE ROLE OF SERIOUS GAMES IN PROFESSIONAL SKILL DEVELOPMENT

Serious games are increasingly recognized as potent tools for cultivating diverse professional skills. They integrate educational objectives with engaging gameplay, creating immersive virtual environments for active learning, problem-solving, and critical thinking. This approach enhances motivation and provides a safe setting for practicing real-world scenarios, improving decision-making and awareness of consequences. This domain was formally established as early as 2005 by Michael and Chen ([8]) as games designed primarily for purposes other than pure entertainment, with subsequent systematic reviews over the past decade focusing on the necessity of rigorous evaluation procedures, including pre/post-tests (Boyle et al. [9]), the integration of real-time affective analytics and virtual reality technologies (Antoniou et al. [10]), the content evaluation of web-based virtual learning environments (Pereira et al. [11]), and the role of serious games within the framework of Education 4.0 (Brandl and Schrader [12]).

The application of serious games in professional development is widespread. Healthcare uses them for risk-free clinical skills training through simulated medical scenarios (Pistono [13]). Military organizations employ them for realistic strategic planning and teamwork simulations (Lamb [14]). Additionally, games enhance social awareness in social educators and develop sustainability-related professional skills, aligning with UN Sustainable Development Goals (Miguel et al. [15]).

Despite their benefits, deploying serious games presents challenges. Assessing their learning effectiveness remains difficult, prompting frameworks like the Framework for Adaptive Serious Games (F4ASG) to guide implementation and assessment (Pistono et al. [16]). Effective design is also crucial: game mechanics must align with pedagogical objectives for meaningful learning (Lamb [14]). Thus, while offering innovative learning opportunities, the success of

serious games depends on careful design and implementation, highlighting the need for ongoing research in this evolving field.

C. DIGITAL HUMANS AND VIRTUAL AVATARS IN EDUCATIONAL SIMULATIONS

Integrating digital humans and virtual avatars into educational simulations enhances interactivity, personalization, and learner engagement. These digital entities serve diverse pedagogical roles, from adaptive teaching assistants to facilitators of immersive learning, reshaping knowledge conveyance.

Digital humans and avatars act as virtual teaching assistants, offering personalized support tailored to individual learning needs and styles (Zhu [17]). They also enable interactive learning through role-playing and simulations, allowing learners to engage in realistic scenarios that promote critical thinking and problem-solving (Wanless-Sobel [18]). Avatars exhibiting emotions and social cues can foster more relatable and engaging learning environments, potentially improving learning and retention (Swartout et al. [19]).

The impact of digital representations on learning outcomes is under active research. Avatar design and type can significantly influence learner performance and engagement. For example, Kao and Harrell [20] noted that while avatars are widely used, including in STEM education for dynamic interactions, highly customized avatars sometimes correlated with lower performance than neutral ones. Despite their pedagogical potential, widespread adoption of digital humans faces challenges like technological requirements, development costs, and ethical considerations (Wanless-Sobel [18]).

D. LARGE LANGUAGE MODELS (LLMS) FOR ENHANCED EDUCATIONAL INTERACTION

LLMs are increasingly integrated into education and training to augment feedback and enrich dialogue-based learning. LLMs' natural language processing capabilities facilitate personalized engagements, making them valuable tools for improving learning outcomes across various educational areas.

A key application is personalized feedback. Tools like Feedback Copilot use LLMs to provide tailored critiques on student assignments, addressing individual needs (Pozdniakov et al. [21]). LLMs also enable sophisticated analysis of educational data; Parker et al. [22] show they can analyze feedback surveys to identify curricular gaps or evaluate teaching effectiveness via thematic and sentiment analysis, sometimes achieving near human-level performance without extensive labeled data.

LLMs also enhance broader learning experiences. Their ability to understand and generate human-like text supports adaptive learning, customizing content and delivery based on student preferences and performance for a more engaging environment. Additionally, LLMs assist in content generation (e.g., quizzes, summaries), streamlining teaching and providing diverse student resources (Dong et al. [23]).

Despite LLMs' advantages in education, their deployment faces critical challenges, including ethical implications, data privacy, and the potential to undermine students' critical thinking if not implemented thoughtfully (Dong et al. [23]). Addressing these is crucial for their effective and responsible integration.

LLMs are progressively being integrated into diverse educational and training contexts, primarily to augment feedback mechanisms and enrich dialogue-based learning interactions. The inherent natural language processing capabilities of LLMs facilitate highly personalized engagements, positioning them as valuable instruments for potentially improving learning outcomes. Their application is evident across several key areas within the educational landscape.

E. SPEECH RECOGNITION IN INTERACTIVE LEARNING ENVIRONMENTS: BENEFITS AND CHALLENGES

Integrating Automatic Speech Recognition (ASR) into interactive learning environments presents both considerable advantages and significant operational challenges. While potentially enhancing learner engagement and personalization, ASR systems face limitations in accuracy and adaptability, especially in complex acoustic and linguistic conditions.

Key challenges include ASR accuracy, as systems often struggle with natural conversational data, leading to high word error rates, particularly in multilingual contexts or with diverse accents (Liesenfeld et al. [24]). Real-time interaction dynamics, such as overlapping speech, also complicate transcription and intent understanding. Moreover, ASR systems exhibit higher error rates than human transcribers, potentially hindering communication and learning (Pucci [25]).

Conversely, speech recognition in educational tools offers notable benefits. It enables personalized learning, with algorithms tailoring exercises to individual proficiency and providing immediate feedback on comprehension or pronunciation. This interactivity enhances learner engagement, as ASR platforms promote active student participation, improving teaching and learning quality. Real-time feedback capabilities, where systems offer instant evaluations and suggestions, also foster dynamic learning (Xue [26]).

In summary, while ASR advancements expand its utility in interactive learning, effectiveness depends on addressing its challenges. Ongoing research is vital to improve ASR systems' reliability, robustness, and user experience in varied educational applications.

F. SYNTHESIS, IDENTIFIED RESEARCH GAPS, AND RATIONALE FOR THE CURRENT STUDY

This review highlights significant technological advancements pertinent to professional skills training. Job interview preparation (Section II-A) increasingly uses AI for feedback and simulation. Serious games (Section II-B) show potential for skill development through engaging environments. Digital humans and virtual avatars (Section II-C) promise

greater realism and interactivity, while Large Language Models (LLMs) (Section II-D) enable dynamic dialogue and personalized feedback. Speech recognition (Section II-E), despite challenges, facilitates natural voice interactions.

However, despite individual promise in these areas, a notable gap exists regarding the holistic integration and empirical evaluation of these cutting-edge technologies into a comprehensive, accessible, and highly realistic serious game for graduate job interview training. To substantiate the novelty claim and clearly position the proposed work, Table 1 provides a direct, feature-by-feature comparison between our IIAA system and representative existing interview platforms.

As shown in Table 1, existing tools often lack crucial dimensions:

- Profound realism and nuanced interactivity from state-of-the-art digital humans (e.g., MetaHumans).
- Truly dynamic, unscripted, contextually relevant dialogues and in-depth feedback from advanced LLMs.
- Engaging, motivating practice environments offered by serious game principles, beyond simple Q&A formats.
- Robust handling of specific linguistic contexts, like Thai language and code-switching, affecting speech-driven interaction.

This research addresses these gaps through the design, development, and initial user-based evaluation of the IIAA. The IIAA is a novel serious game synergistically combining a high-fidelity MetaHuman interviewer, an LLM-powered conversational AI for dynamic interaction and feedback, and integrated speech technologies—speech-to-text (STT), text-to-speech (TTS), and speech-to-animation (STA)—within an Unreal Engine immersive environment. This study explores the feasibility of this integration, assesses its effectiveness in creating a realistic and engaging practice experience, and evaluates its utility for enhancing graduate interview preparedness, particularly in the Thai context. Section III (Methodology) details the system's development and evaluation.

III. METHODOLOGY

The system developed in this study is an intelligent interactive environment, engineered through the integration of advanced tools within Unreal Engine 5.2. The architecture of this system is characterized by two principal components:

1. Front-end: This component serves as the primary user interface, facilitating interaction predominantly through voice-based commands.
2. Back-end: This component incorporates a conversational agent (chatbot) that manages real-time communication flows. Its core functionalities include STT conversion, TTS synthesis, speech-driven facial animation for the digital avatar, and the rendering of a realistic digital human.

The subsequent sections provide a detailed exposition of each architectural component. This includes an elaboration on the selection criteria for the employed technologies, their respective integration methodologies, and the optimization strategies implemented to ensure robust and seamless system performance. A schematic overview of the application architecture is presented in Figure 1.

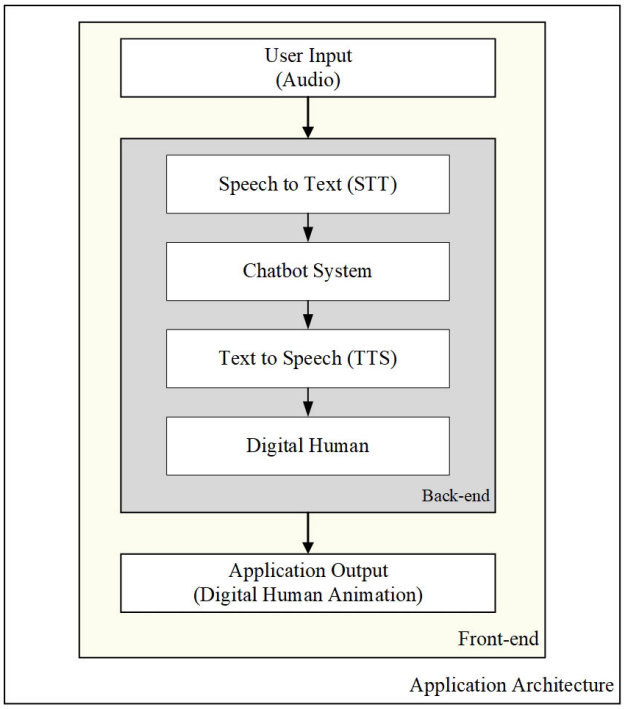

FIGURE 1. Schematic diagram of the overall system architecture.

A. DEVELOPMENT OF THE INTERACTIVE INTERVIEW ASSISTANT APPLICATION (IIAA)

This section details the development process and core technological components of the IIAA, a system designed to simulate job interview scenarios.

1) SPEECH-TO-TEXT (STT) SYSTEM

A primary design consideration for the IIAA was its specific requirement to support the Thai language. Consequently, this subsection outlines the comparative evaluation of three STT models, selected for their documented Thai language support capabilities:

1. **Whisper**: Proposed by Radford et al. [27], the Whisper framework encompasses several model variants. For this study, the 'medium' variant, which comprises 769 million parameters and offers multilingual support including Thai [28], was selected. This model variant provides a balance between high-quality speech recognition performance and computational resource demands, rendering it more efficient for this application compared to its larger counterparts.
2. **Thonburian Whisper (Whisper Medium Thai Combine V.3)**: This model represents a fine-tuned version of Whisper Medium. It was trained utilizing augmented datasets, including Common Voice 13.0 (Thai), Google Fleurs, and other curated Thai language corpora,

TABLE 1. Feature comparison of AI interview platforms.

Feature	IIAA System (Our Work)	EZInterviewer / DuerQues	Generic GPT-based Systems
Interviewer Realism	High-Fidelity MetaHuman (UE5)	2D Image / Basic 3D Avatar	Text-only or Basic TTS Avatar
Dialogue Engine	Advanced LLM (Dynamic, Unscripted)	Primarily Scripted Dialogue Trees	Advanced LLM (Dynamic, Text/TTS)
Speech-to-Animation (STA)	Fully Integrated Real-time Facial/Body Sync	Absent or Pre-canned Animations	Absent (Text-only interaction)
Holistic Integration	LLM + STT + TTS + STA + UE5 (Single Platform)	Modular (Dialogue + Basic TTS/STT)	Modular (LLM API + External Services)
Targeted Linguistic Context	Focus on mixed Thai-English Code-Switching	Generally limited to major single languages (e.g., English)	Varies by STT provider, no Avatar context

specifically to enhance performance in Thai automatic speech recognition (ASR) [29].

3. **wav2vec2-large-xlsr-53-th**: This model [30], specifically fine-tuned for Thai ASR, is derived from the wav2vec2 architecture originally proposed by Alexei et al. [31]. The Thai-specific model was developed by the AI Research Institute of Thailand, a collaborative initiative between Vidyasirimedhi Institute of Science and Technology (VISTEC) and the Digital Economy Promotion Agency (DEPA).

The performance of these STT models was experimentally evaluated using a corpus of five predefined Thai phrases to ascertain their respective processing speeds and recognition accuracies. The test phrases in English translations were as follows:

1. Hello.
2. What are you doing?
3. What should I eat for lunch?
4. I can write in C and Python.
5. Besides programming, I also have skills like 3D modeling and an interest in railway engineering.

To ensure reliability of the performance metrics, each phrase was processed 10 times by each STT model, facilitating the assessment of both processing speed and transcription accuracy. This comparative evaluation aimed to identify the optimal STT model for integration into the IIAA, prioritizing effective and accurate transcription of spoken user input during interactions with the digital human interface.

As presented in Figure 2, the Whisper model exhibited the highest conversion speed (measured in words per second). However, its transcription accuracy was found to be low (Figure 3), rendering it unsuitable for practical application within the IIAA. In contrast, while Thonburian Whisper achieved high transcription accuracy, its processing speed was substantially lower than that of the wav2vec2-large-xlsr-53-th model. Consequently, the wav2vec2-large-xlsr-53-th model was selected as the STT engine for the IIAA. This decision was based on its favorable balance of high processing speed and a high accuracy rate (95.58%), which was deemed sufficient for the operational requirements of the conversational system.

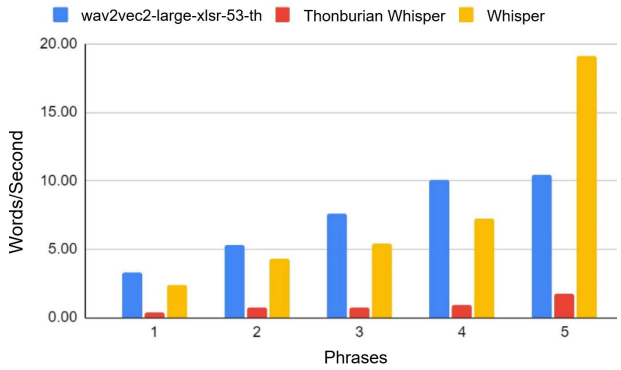

FIGURE 2. Comparative analysis of STT model processing speeds (words per second).

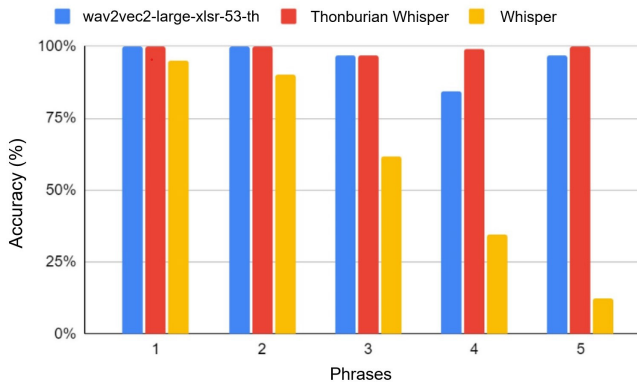

FIGURE 3. Comparative analysis of STT model transcription accuracy.

Within the Unreal Engine environment, audio input is captured using the Runtime Audio Importer plugin [32]. Subsequently, the recorded audio data is transmitted for STT processing to a dedicated Application Programming Interface (API) endpoint on the Hugging Face platform, a process facilitated by the Restful plugin. The transcribed text output from the STT service is then relayed to the core conversational AI module (detailed in Section III-A2). This integrated pipeline is designed to ensure efficient and accurate STT conversion, forming a critical component of the chatbot's input mechanism. The overall STT process flow is illustrated in Figure 4.

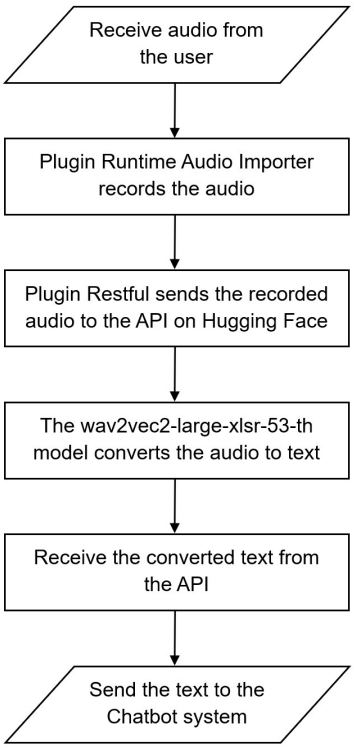

FIGURE 4. Process flow diagram of the STT integration.

2) LARGE LANGUAGE MODELS (LLM) FOR CONVERSATIONAL AI

The core of the IIAA's conversational capabilities is an LLM. The selection and integration process for the LLM is detailed below.

a: LANGUAGE MODEL SELECTION

Two prominent LLMs were initially considered for integration within the Unreal Engine environment: LLaMA 2 and ChatGPT 3.5. In our specific implementation and testing environment, preliminary experimental testing within Unreal Engine indicated that LLaMA 2 exhibited instability, leading to frequent engine crashes. Furthermore, its substantial storage requirement of approximately 60 GB rendered it impractical for the intended application deployment. Consequently, ChatGPT 3.5 was selected for more thorough evaluation. This model is offered in two primary variants: ChatGPT 3.5 Turbo 4k and ChatGPT 3.5 Turbo 16k. The ChatGPT 3.5 Turbo 16k variant, characterized by a larger context window and parameter size, demonstrated superior conversational response capabilities. It was therefore chosen for final integration due to its operational stability, performance characteristics, and manageable resource footprint.

The selected ChatGPT 3.5 Turbo 16k model is accessed via its official Application API, facilitated within Unreal Engine by the Http GPT Plugin. This integration enables the transmission of user inputs (as text from the STT system) to the ChatGPT API. The LLM processes these inputs based on structured prompts (detailed in Section III-A2b) and generates a textual response. This response is then returned to the Unreal Engine system, thereby enhancing the interactive capabilities of the simulated interview.

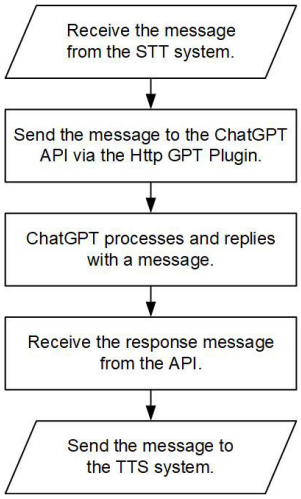

FIGURE 5. Chatbot operational workflow illustrating the integration of STT, ChatGPT (via API), and TTS components within Unreal Engine.

The overall chatbot workflow, illustrating the integration between Unreal Engine and ChatGPT, is depicted in Figure 5. The process commences with the STT system (Section III-A1) converting spoken user input into text. This text is subsequently transmitted to the ChatGPT API using the Http GPT Plugin. Upon receiving the processed textual response from ChatGPT, the system forwards this to the TTS engine (Section III-A3) for conversion into audible speech for the user. This orchestrated workflow ensures a coherent transition from voice input to an audible, AI-generated response.

b: PROMPT ENGINEERING FOR INTERVIEW SIMULATION AND EVALUATION

To enable ChatGPT to function as a simulated Human Resources (HR) professional within the interview system, a meticulously structured prompt was designed. This prompt established the LLM's persona, operational guidelines, and evaluation responsibilities for the specified job interview context.

The core components of the primary prompt included:

- **Role Persona Definition**: The LLM was instructed to adopt the persona of an HR professional possessing expertise in the selected job domain. The directive was: "From now on, you are an HR professional with advanced knowledge about *job_title*, and you will interview me for a position in *job_title*." The *job_title* placeholder was dynamically populated based on user selection.
- **Behavioral Guidelines**: A set of rules was provided to govern the LLM's interactional behavior:
  - Input Ambiguity Resolution: "If I speak unclearly, please infer my intended meaning." This instruction aimed to enhance conversational robustness.

- Topic Adherence: "If I discuss topics outside the interview, respond with 'Sorry, I cannot answer that question.'" This was implemented to maintain focus on the interview agenda.
- Cultural Formality (Thai Language)[1]: "Use 'khráp' at the end of sentences if your persona is male, and 'khâ' if female." This ensured appropriate gender-specific polite particles in Thai.
- Permissible Opinion Expression: "You may offer slight opinions on answers if appropriate." This allowed for nuanced, human-like interaction.
- Interactional Realism: "Conduct the interview as if it were a real one." This guided the LLM towards emulating authentic interview dynamics.
- Interview Initiation Protocol: "Begin the interview immediately upon receiving the user's name."

For the assessment of candidate (user) performance, ChatGPT was further instructed to act as an evaluating HR interviewer. Its task was to evaluate the user's responses to interview questions on a 10-point scale, considering both relevance and appropriateness. This quantitative score, accompanied by qualitative textual feedback, was systematically recorded for subsequent post-interview analysis by the user. The LLM-driven evaluation process encompassed the following elements:

- **HR Perspective Simulation**: The LLM assigned scores to user responses by emulating the evaluative criteria an HR professional might employ.
- **Dialogue Structuring**: The conversational output was formatted to clearly differentiate between the interviewer's (ChatGPT) utterances and the candidate's (user's) statements, ensuring clarity in the interaction log.
- **Comprehensive Feedback Provision**: For each user response, ChatGPT generated both a numerical score and a concise written critique. This dual-feedback mechanism was designed to facilitate detailed data analysis and user reflection within the Unreal Engine interface.

This systematic approach to prompt engineering and LLM instruction was implemented to enable consistent and efficient candidate assessment, providing users with structured, actionable feedback for their post-interview evaluation and learning.

c: DYNAMIC INTERVIEW QUESTION GENERATION

To enhance the interactivity and contextual relevance of the simulated interview, a dynamic question generation system was implemented. This system enabled the LLM (ChatGPT) to pose questions pertinent to the user's selected career path. Following each user response, the system was programmed to randomly select subsequent questions from a predefined, career-specific question bank and instruct ChatGPT to present it. This question generation mechanism featured two distinct operational modes, contingent on the desired interview depth: 'basic' and 'comprehensive'.

In the 'basic' interview mode, the system iteratively selected and posed questions from the relevant bank in a randomized sequence until the predetermined set of questions for a standard interview was exhausted. For the 'comprehensive' interview mode, a more adaptive approach was employed. After an initial question on a specific topic was posed and the user had responded, ChatGPT was prompted to explore that topic in greater detail. Following each subsequent user response within this topical exploration, the system incorporated a probabilistic decision point: there was a 50% probability that ChatGPT would be instructed to continue delving into the current topic with a follow-up question, and a 50% probability that it would introduce a new question from a different topic area. This iterative process of topical exploration and new question introduction continued until all designated topics were covered, at which juncture a specific prompt directed ChatGPT to conclude the interview.

Upon completion of either interview mode, a summary interface was presented to the user, displaying the assigned scores and qualitative critiques for each segment of the dialogue. This structured, dynamic approach to question management was designed to ensure that the simulated interview remained engaging and adaptable to the user's preferred level of detail, while culminating in a thorough performance evaluation. The logic of this dynamic question system is illustrated in Figure 6.

d: CAREER PATH SPECIALIZATION AND QUESTION BANK DESIGN

The IIAA allows users to select one of three distinct career paths for their simulated interview: UI/UX Designer, Software Engineer, or Game Developer. These specializations correspond to undergraduate programs offered at School of Information Technology, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand. For each career path, a dedicated bank of interview questions was curated. These questions were derived from an analysis of authentic job descriptions sourced from reputable industry recruitment platforms. The design of these question prompts was meticulously undertaken to assess a candidate's specific qualifications, technical competencies, and soft skills pertinent to their chosen profession.

The thematic focus of the questions for each career path is exemplified below:

- **UI/UX Designer**: Candidates are evaluated on their comprehension of user-centered design principles, proficiency with industry-standard design software (e.g., Figma, Adobe XD), and collaborative abilities within a team setting. Question topics also encompass prototype

[1]Thai language includes gender-specific particles used to indicate the speaker's gender and convey politeness. Males use ***khráp***, while females use ***khâ***. These particles are key elements in formal speech and are studied in sociolinguistics as part of gendered language.

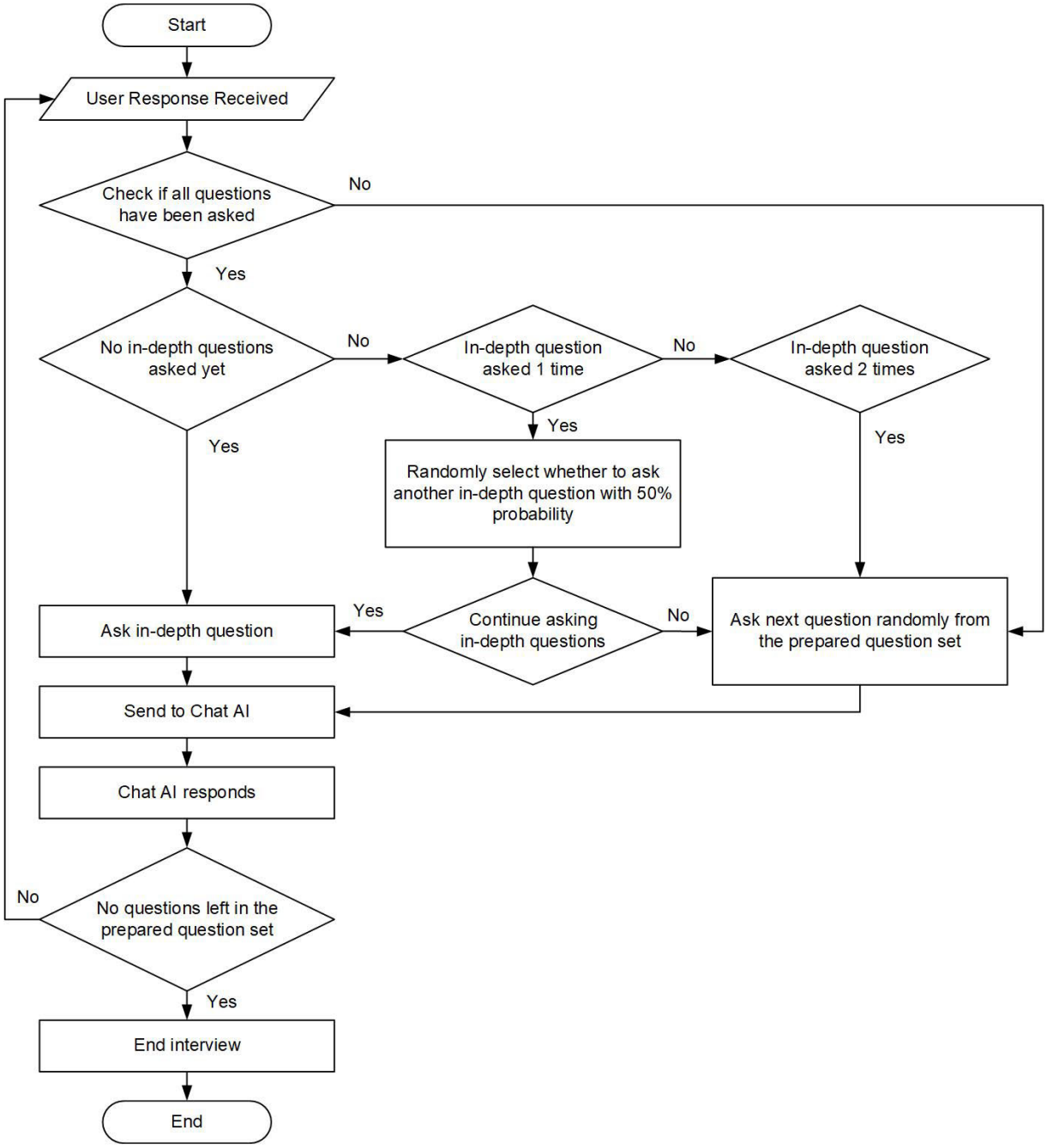

FIGURE 6. Flow diagram illustrating the logic of the dynamic question generation system for basic and comprehensive interview modes.

development methodologies, wireframing techniques, and user research practices to provide a holistic assessment of their design capabilities.

- **Software Engineer**: Assessment criteria for this role include knowledge of database management systems (DBMS), demonstrated teamwork and problem-solving skills, and proficiency in software troubleshooting and debugging. Further questions explore their adaptability to new technologies and practical experience with relevant programming languages (e.g., Java, PHP, JavaScript, Python, C++).
- **Game Developer**: Candidates are assessed on their programming proficiency (e.g., C++, C#), experience with contemporary game engines (e.g., Unreal Engine, Unity), and their analytical skills related to game design principles. The questions also aim to gauge their commitment to the field of game development and their capacity for effective teamwork in a development environment.

This structured and specialized approach to question bank development ensures that the LLM-driven interviewer delivers a targeted and relevant evaluative experience, thereby

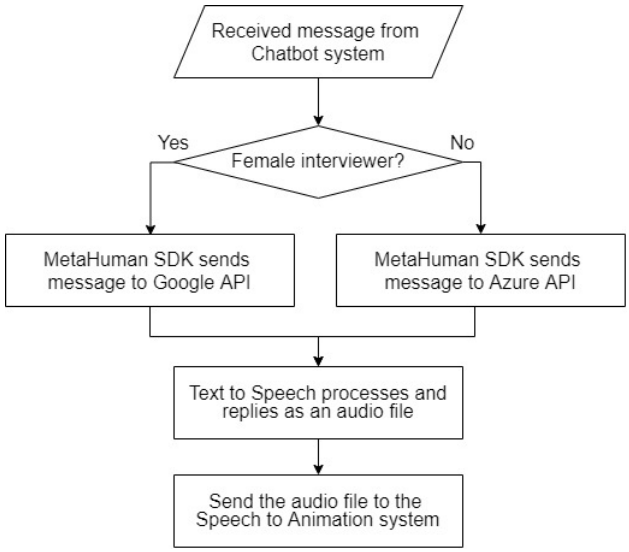

FIGURE 7. Flowchart illustrating the TTS generation and integration process.

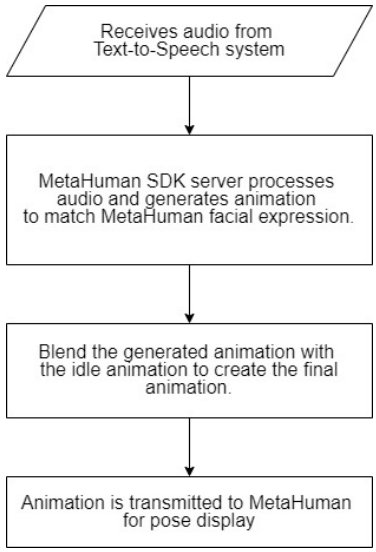

FIGURE 8. Flowchart detailing the STA process, including integration of supplementary idle animations.

enhancing the efficacy and specificity of the interview simulation for each career track.

3) TEXT-TO-SPEECH (TTS) IMPLEMENTATION

To imbue the digital human interviewer with vocal capabilities, textual responses generated by the LLM (as described in Section III-A2) were converted into audible speech using TTS technology. This was facilitated by the MetaHumanSDK plugin within Unreal Engine, which provides an integrated suite of tools for creating realistic digital human interactions, including TTS functionalities from providers such as Google Cloud and Microsoft Azure.

The selection of the specific TTS engine and voice was customized based on the preselected gender of the interviewer avatar to enhance persona consistency. For female avatars, Google Cloud's Text-to-Speech engine was employed, utilizing the Thai voice ID th-TH-Neural2-C. For male avatars, Microsoft Azure's Text-to-Speech service was implemented, with the Thai voice ID th-TH-NiwatNeural. These specific neural voices were chosen for their naturalness and alignment with the respective avatar genders.

The audio output generated by the TTS engine was subsequently synchronized with the MetaHuman facial animations (detailed further in Section III-A4) to ensure audiovisual coherence during the simulated interview. This integration of gender-specific, natural-sounding TTS with realistic avatar animation was intended to augment the perceived realism of the virtual interviewer, thereby contributing to user engagement and the overall authenticity of the simulation. The process flow for the TTS implementation is depicted in Figure 7.

4) SPEECH-TO-ANIMATION (STA) FOR AVATAR REALISM

Subsequent to the generation of audible speech by the TTS engines (as detailed in Section III-A3), the resultant audio data was processed to generate synchronized facial animations for the selected MetaHuman avatar. This STA conversion was managed via the MetaHumanSDK, which transmits the audio to a dedicated cloud server. The server then returns animation data designed to drive the avatar's facial musculature in alignment with the articulated speech, thereby enhancing the perceived realism of the digital human by simulating natural vocal delivery.

Initial testing of the STA pipeline revealed a limitation: the default animations were predominantly restricted to lip synchronization and basic mouth movements. This resulted in the MetaHuman avatar appearing somewhat static and lacking in natural expressiveness. To mitigate this, supplementary idle animations were sourced from general animation asset libraries and integrated into the system. These assets provided a broader range of subtle, dynamic facial expressions (e.g., blinks, minor brow movements) that were then layered with the speech-driven animations to reduce the static appearance. Furthermore, dedicated idle body animations (e.g., slight shifts in posture, subtle torso movements) were developed and synchronized with both the facial speech animations and the supplementary facial idle animations.

The integration of these multi-layered animation components —core speech-driven facial movements, supplementary facial idle expressions, and holistic body idle animations— significantly improved the perceived naturalness and dynamism of the MetaHuman avatar. This comprehensive approach to STA was crucial for enhancing the overall believability and engagement of the digital human, ensuring more lifelike movements and expressions throughout the user interaction. The STA process flow is illustrated in Figure 8.

5) USER INTERFACE (UI) AND USER EXPERIENCE (UX) DESIGN

The design of the User Interface (UI) and User Experience (UX) for the IIAA was guided by principles of clarity,

FIGURE 9. User interface design of the start page.

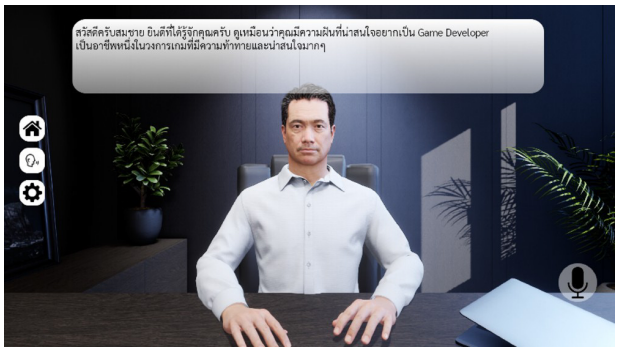

FIGURE 10. User interface design of the main interview page.

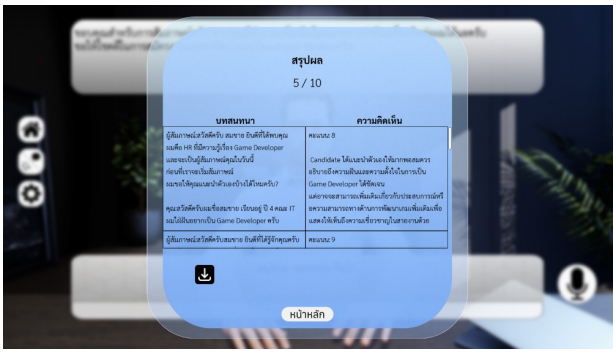

FIGURE 11. User interface design of the post-interview summary page.

efficiency, and intuitive navigation to facilitate a seamless interview simulation process. The primary objective was to develop a user-friendly environment enabling participants to easily input requisite information, interact with the conversational agent, and review their performance outcomes. The design prioritized simplicity and functionality across three key interfaces: the Start Page, the Interview Page, and the Summary Page. All UI elements were developed with Thai language localization to ensure accessibility for the target user group.

The specific design and functionalities of each interface are detailed below:

- **Start Page** (Figure 9): This initial interface serves as the entry point to the application. It prompts users to input their name and select one of the three available job profiles (UI/UX Designer, Software Engineer, or Game Developer). Navigational elements include a 'Next' button, which directs the user to an avatar and environment customization screen (not detailed here but implied as next step), and an 'Exit' button to terminate the application.
- **Interview Page** (Figure 10): This is the primary interaction screen where the simulated interview takes place. It features a dedicated area at the top for displaying the interviewer's (chatbot's) responses and a larger section below for the ongoing conversation history. Users can input their responses via a text field or by activating a microphone icon for voice input. A persistent 'Home' button allows users to return to the Start Page. Additional controls on the right-hand side provide options to toggle the visibility of the text input box and access a settings/configuration panel.
- **Summary Page** (Figure 11): Upon conclusion of the interview simulation, users are navigated to this page. It provides a consolidated overview of their performance, including average scores and a detailed log of the conversation with specific scores and qualitative comments appended to each dialogue exchange. A 'Download' function is available, enabling users to save their interview history for offline review. A 'Home' button facilitates a return to the Start Page.

6) OPTIMIZATION STRATEGIES

This subsection details specific strategies implemented to address two key performance challenges identified within the IIAA: latency in TTS processing and initialization delays affecting STT model availability. The objective of these optimizations was to reduce system response times, thereby ensuring smoother user interactions and an improved overall user experience.

- **Optimization of Text-to-Speech (TTS) Response Pacing:**

Initial observations revealed that perceived delays in chatbot responses were often attributable to the TTS system awaiting complete sentence generation from the LLM (ChatGPT) before initiating speech synthesis. Furthermore, a naive approach of segmenting text for TTS processing at every whitespace was found to introduce excessive, unnatural pauses. To mitigate these issues, an experiment was conducted to determine an optimal minimum character threshold for segmenting and submitting text to the TTS engine post-whitespace detection. This experiment evaluated the trade-off between the initial delay before speech onset and the cumulative pause duration between synthesized speech segments.

The results of this character-splitting performance evaluation are presented in Table 2. An increasing minimum character count for TTS submission led to longer initial delays before the avatar commenced speaking, due to the

increased processing time required for larger text chunks. Conversely, a lower character threshold resulted in more frequent, smaller submissions to the TTS system. While this reduced initial latency, it increased the inter-sentence pause durations, leading to a disjointed conversational flow. Based on these empirical findings, a minimum character limit of 75 characters was selected. This threshold was determined to provide the best balance between prompt speech initiation and natural conversational pacing by minimizing the total duration of chatbot inactivity.

TABLE 2. Performance metrics for TTS processing based on varying minimum character thresholds for speech synthesis initiation.

Minimum Characters for TTS (chars)	Delay Before Speech (sec)	Pause Between Sentences (sec)	Total Chatbot Inactivity (sec)
0	3.97	71.70	75.67
25	4.41	61.12	65.53
50	5.60	36.42	42.02
75	**6.81**	**24.82**	**31.63**
100	9.95	22.11	32.06

- **Mitigation of Speech-to-Text (STT) Model Cold-Start Latency:**

A notable operational challenge encountered with the STT model (hosted on the Hugging Face API infrastructure) was its delayed readiness due to 'cold starts'. This phenomenon refers to an initial activation period required by the serverless infrastructure to load the model into memory upon first request after a period of inactivity, thereby causing significant delays during the initial use in an interview session. To mitigate this, a preemptive initialization strategy was implemented. This involved programmatically sending an initial request to the STT API endpoint before the user commenced their first interaction. This "wake-up" call ensures that the model is loaded and ready for immediate processing when the user begins to speak. In instances where the model might still not be fully active, the system is designed to make supplementary preemptive requests to minimize any residual delay. This proactive solution substantially enhances the STT model's responsiveness from the outset of the interaction, thereby improving the user experience by reducing initial waiting times.

B. ETHICAL CONSIDERATIONS AND DATA PRIVACY

The study adhered strictly to institutional ethical guidelines for research involving human participants. Informed consent was obtained from all participants prior to their involvement in the study.

Crucially, we implemented a real-time, non-persisting data model to protect user privacy. All user interactions, including voice input and text responses, were processed in real-time via external API calls (STT, LLM) and were not stored on any permanent server associated with the IIAA system or the institutional research infrastructure. Specifically:

- Voice Data: User voice recordings were transmitted to the STT API solely for transcription and were immediately discarded post-processing.
- Chat Logs and Responses: Conversational history, while generated by the LLM, was treated as transient data and was not logged or stored in a manner that could be linked back to the participant's identity.
- Performance Scores: Final, anonymized performance scores and system usage metrics were stored only after being de-identified and aggregated for analysis purposes.

This methodology ensured that no user-identifiable voice recordings or chat logs were maintained, mitigating risks related to data breach, unauthorized access, and reproducibility issues stemming from persistent data storage.

C. USER EVALUATION STUDY

To assess the efficacy and usability of the developed IIAA, a user evaluation study was conducted. This section details the participant cohort, the experimental procedure, and the data collection methods employed.

1) PARTICIPANTS

The user study involved 10 final-year undergraduate students from School of Information Technology, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand. Participants were recruited from distinct areas of specialization relevant to the career paths simulated by the IIAA: five majoring in UX/UI Design, three in Software Engineering, and two in Game Development. All participants had proficiency in the Thai language, the primary language of interaction for the system. Table 3 details the specific demographics and specialization areas of the participant cohort.

TABLE 3. Participant demographics (N = 10).

Demographic Category	Count
Total Participants (N)	10
Academic Standing	Final-Year Undergraduate
Primary Language of Interaction	Thai
Major / Simulated Career Path	
UX/UI Design	5
Software Engineering	3
Game Development	2

2) PROCEDURE AND TASK

The experimental procedure commenced with an introductory briefing, after which participants provided informed consent. Each participant was then guided to enter basic personal information (e.g., name) into the system and subsequently select their preferred job role for the simulated interview from the available options: UX/UI Designer, Software Engineer, or Game Developer. Participants were also afforded options to customize the interview environment, which included selecting the interviewer's avatar gender (male or female) and choosing one of three virtual office settings. Furthermore, participants selected their desired interview format: either a 'basic' session, with an approximate duration of 10 minutes,

or a 'comprehensive' session, extending to approximately 30 minutes.

Upon commencement of the interview, the AI-driven interviewer, configured according to the participant's selected career path and interview type, initiated the dialogue. Participants could submit their responses using either voice input (via microphone) or keyboard text entry. A key feature of the interface allowed for the review and potential correction of responses prior to final submission. This was particularly emphasized for voice interactions to mitigate potential inaccuracies arising from STT conversion, especially given the common linguistic phenomenon of Thai-English code-switching in colloquial Thai. Following the interview's conclusion, the system presented participants with individualized performance feedback, which included numerical scores for each question answered and qualitative comments generated by the LLM.

3) DATA COLLECTION AND MEASURES

Data for the evaluation of the IIAA were systematically collected across several categories:

- System Performance Metrics: Processing speeds for the core system components (STT, ChatGPT, TTS, and STA), as detailed in Section III-A, were measured in characters per second (cps). While these are primarily system-centric metrics, they provide context for the user experience, particularly regarding responsiveness. (The detailed results of these measurements are presented in Section IV-A).
- Interview Feedback Quality: The LLM (ChatGPT) was programmed (as described in Section III-A2b) to provide qualitative feedback and assign quantitative scores (on a 1-10 scale) for each user response, based on perceived relevance and depth. This AI-generated feedback was collected as a measure of the system's evaluative output.
- User-Perceived Usability: System usability was quantitatively assessed using a survey instrument adapted from the System Usability Scale (SUS) [33]. To allow for more granular participant feedback, the standard 5-point Likert scale typically associated with SUS items was expanded to a 10-point scale. This modification was informed by usability research suggesting potentially increased sensitivity with finer rating scales [34].
- Aesthetic and Utility Perceptions: In addition to the SUS-based items, specific participant ratings were solicited (using a 10-point scale) concerning perceptions of system aesthetics (e.g., interface design attractiveness, visual appeal of the avatar) and various facets of utility. Utility-focused questions addressed the perceived accuracy of speech detection, clarity of the chatbot's synthesized speech, relevance of interview questions for effective preparation, usefulness of the post-interview feedback provided by the system, and the system's impact on participants' self-reported confidence levels for actual job interviews.

IV. RESULTS

This section presents the findings derived from the system performance evaluation, analysis of AI-generated interview feedback, and user satisfaction assessments. The performance of key architectural components—STT, ChatGPT, TTS, and STA—was quantitatively assessed. Interview feedback, generated by the LLM, was evaluated based on scores assigned on a 1-10 scale reflecting response depth and relevance. User satisfaction was measured across several criteria, including system accuracy, clarity of communication, and perceived usefulness. The subsequent subsections detail these findings, offering insights into the system's operational effectiveness.

A. SYSTEM PERFORMANCE

The performance of four key system components was evaluated: STT, ChatGPT, TTS, and STA. Given their shared library integration (MetaHumanSDK), the TTS and STA components were assessed as a combined process. Performance metrics, presented in Table 4, reveal that the combined TTS + STA process constitutes the primary system bottleneck, operating at a mean rate of 21.09 characters per second (cps). This significantly impacted overall system responsiveness. In contrast, the STT and ChatGPT components demonstrated higher efficiency, processing at mean rates of 50.38 cps and 45.64 cps, respectively. The observed latency in the STA component is attributed to the computational demands of real-time rendering of high-definition 3D avatars, a necessary feature for the intended realistic interview simulation.

TABLE 4. Mean processing speed (characters per second) of core system components.

STT	ChatGPT	TTS + STA
50.38	45.64	21.09

B. INTERVIEW FEEDBACK ANALYSIS

The quality of interview feedback, generated by ChatGPT based on predefined prompt structures (as described in Section III-A2), was analyzed. User responses were scored by the LLM on a 1-10 scale, reflecting perceived depth and relevance to the interview questions. The following examples illustrate the typical scoring rationale across different performance tiers:

Low Score (1-4)

Q: How extensive is your knowledge and understanding of game development principles and various game components, and do you have knowledge of game design and resource management within games?

A: I have a decent understanding of MDAs (Mechanics, Dynamics, Aesthetics) from my experience playing lots of games.

Score: 4

Feedback: The response demonstrates a basic grasp of game development principles but lacks detail and depth. While the applicant's mention of MDAs shows some

familiarity, the answer is insufficiently detailed regarding game design and resource management. To enhance the response, the applicant should include specific examples and elaborate on their knowledge and experience, thereby improving the perceived expertise.

Medium Score (5-7)

Q: Using Unreal Engine is very interesting. How long have you been developing games with Unreal Engine? Do you have any projects or works developed with Unreal Engine that you would like to share?

A: I have been developing games with Unreal Engine for two years. I also have a project that is a horror game combined with puzzle elements.

Score: 6

Feedback: The response reflects relevant experience and involvement with Unreal Engine, indicating a significant project. However, the feedback suggests that the answer could be more informative. Providing additional details about the project, such as its scope, challenges, and outcomes, would offer a clearer understanding of the applicant's skills and contributions, thereby improving the overall assessment.

High Score (8-10)

Q: Regarding teamwork skills, do you have experience working as part of a team, and are you capable of collaborating with others and solving problems within a team?

A: I have experience working as part of a team. I participate in brainstorming sessions and exchange knowledge with my colleagues. When problems arise, I bring up the issues I have encountered and work together with my teammates to solve them.

Score: 8

Feedback: The response is well-rounded and provides a comprehensive view of teamwork and problem-solving skills. The applicant's description of collaborative efforts and problem resolution demonstrates strong interpersonal and problem-solving capabilities. This detailed response aligns with high performance in teamwork scenarios, reflecting the applicant's suitability for collaborative work environments.

C. USABILITY AND USER SATISFACTION

User-perceived usability and satisfaction were quantitatively assessed using a 10-point rating scale for questions adapted from the SUS framework, as detailed in Section III-C3 The evaluation encompassed three categories: Aesthetic Evaluation, Utility Evaluation, and Overall Experience. Detailed mean scores are presented in Table 5.

For the Aesthetic Evaluation category, mean scores for 'System Ease of Use' and 'Interface Aesthetics' were 7.9 and 8.4, respectively (on the 10-point scale). In the Utility Evaluation category, mean scores for specific aspects were as follows: 'Accuracy of Speech Detection' (7.0), 'Clarity of the Chatbot's Speech' (8.3), 'How Well the Chatbot's Questions Help in Preparing for the Interview' (8.0), 'Usefulness of the Post-Interview Feedback' (8.9), and 'Increased Confidence in Job Interviews After Use' (8.0). The 'Overall Experience' with the system received a mean score of 8.2. The 'Accuracy of Speech Detection' item received the lowest mean score (7.0) within the utility category.

V. DISCUSSION

This section interprets the findings presented in Section IV, contextualizes them within the existing literature, discusses the principal contributions and limitations of the current research, and proposes avenues for future work.

A. KEY FINDINGS AND IMPLICATIONS

The findings of this research underscore the potential of integrating advanced technologies, specifically within a serious game framework, to enhance interview preparation for graduating students. The utilization of MetaHumanSDK, facilitating realistic TTS and STA functionalities, contributed to a more immersive and authentic simulated interview experience.

Although performance metrics revealed that the TTS and STA components, burdened by high-definition 3D model rendering, constrained overall system throughput (as detailed in Section IV-A), the developed application was generally perceived by users to maintain a reasonable balance between interactional responsiveness and avatar realism. Qualitative and quantitative user feedback (Section IV-C) affirmed the system's utility in enhancing participants' perceived interview readiness and offered valuable insights for future iterative development. The AI-generated feedback, based on structured LLM prompts, was also a key component, providing users with specific points for improvement (examples in Section IV-B).

B. LIMITATIONS

Several limitations warrant consideration. Firstly, observed variability in system performance, predominantly associated with the TTS and STA components (as detailed in Section IV-A), is a key concern. The substantial computational overhead required for real-time avatar animation resulted in increased response latency, potentially diminishing aspects of the user experience.

Furthermore, the efficacy of voice-based interaction was partly compromised by suboptimal speech detection accuracy with mixed Thai-English linguistic input (Section IV-C). This highlights a specific challenge for voice interactive systems in multilingual contexts that utilize code-switching.

A third significant limitation arises from the system's dependency on continuous access to real-time online backend services (e.g., LLM and STT APIs). Susceptibility to external service interruptions or outages, which are beyond the direct control of the developed application, poses a risk to consistent system availability and performance.

A fourth limitation relates to the assessment component of the study. The interview performance feedback provided to participants relied solely on automated evaluation generated by the LLM. This methodology lacks validation from human subject-matter experts (e.g., HR professionals or

TABLE 5. Mean user evaluation scores (10-point scale) for system usability and satisfaction across aesthetic, utility, and overall experience categories.

Category	Questions	UX/UI					Software Eng			Game Dev		Ave
		#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	
Aesthetic Evaluation	1) System Ease of Use	10	10	5	8	6	8	7	10	5	10	7.9
	2) Interface Aesthetics	7	10	10	8	7	8	6	10	8	10	8.4
Utility Evaluation	3) Accuracy of Speech Detection	7	10	7	7	3	6	5	10	5	10	7.0
	4) Clarity of the Chatbot's Speech	8	10	10	7	4	8	10	10	8	8	8.3
	5) How Well the Chatbot's Questions Help in Preparing for the Interview	10	10	10	7	4	7	7	8	7	10	8.0
	6) Usefulness of the Post-Interview Feedback	10	10	10	8	7	9	10	8	7	10	8.9
	7) Increased Confidence in Job Interviews After Use	7	10	10	8	2	9	7	10	7	10	8.0
Overall Experience	8) Overall Experience	7	10	10	8	7	8	7	10	5	10	8.2

career counselors), which could introduce potential biases or inaccuracies in the feedback provided and limit the trustworthiness of the LLM's automated scoring.

Finally, the generalizability of our findings is limited by the study's scope, which included a small sample size and was conducted within a single institutional context. These factors constrain the extent to which the results can be extrapolated to different user demographics or diverse educational settings. A critical direction for future research will be to address this by conducting large-scale, multi-institutional studies with a more diverse participant pool to validate and broaden the applicability of the system's efficacy.

C. COMPARISON WITH PREVIOUS RESEARCH

As reviewed in Section II, prior research details advancements in individual technologies relevant to skills training, including AI for interview aids (Section II-A, e.g., [2], [3], [4], [5], [6], [7]) serious games (Section II-B, e.g., [13], [14], [15], [16]) engaging digital humans (Section II-C, e.g., [17], [18], [19], [20]), and LLMs in education (Section II-D, e.g., [21], [22], [23]), and, and speech recognition for interactive learning (Section II-E, e.g., [24], [25], [26]). However, our literature synthesis (Section II-F) identified a key gap: the lack of holistic integration of these state-of-the-art components into a comprehensive serious game for realistic job interview training, with many existing tools focusing on isolated features or lacking advanced interactive realism.

The IIAA directly addresses this gap. It uniquely integrates a high-fidelity MetaHuman interviewer, LLM-powered conversational AI, and speech technologies within an immersive serious game framework. This synergistic design aims for significantly enhanced realism and interactivity. Initial findings suggest this comprehensive approach, despite technical challenges, offers a promising advancement over existing interview preparation tools and contributes to impactful game-based learning and applied AI solutions in education.

D. FUTURE RESEARCH

Future research trajectories should prioritize the optimization of system performance, particularly in enhancing the robustness of STT for complex linguistic inputs (e.g., code-switching) and improving the efficiency of real-time animation rendering.

A crucial priority for future development is the mitigation of the system's dependency on continuous online APIs. This will be addressed by designing a hybrid processing architecture. Key components, such as localized STT processing for brief utterances, basic avatar state machine logic, and initial LLM instruction sets, will be offloaded to the local device. This hybrid approach is projected to reduce external API latency, minimize the risk of external service outages, and enable critical partial offline operation, thereby significantly boosting system reliability and accessibility for end-users.

Crucially, a key objective for future work will be to conduct a rigorous follow-up study to address the generalizability limitations of the current feasibility research and, most importantly, to measure actual learning gains. This subsequent study will utilize a pre-test/post-test design employing a rubric-based performance scoring system to quantitatively measure improvements in interview competency after using the application. It will be designed with a significantly larger, more diverse, multi-institutional participant sample, and will incorporate a control group (e.g., traditional human-to-human mock interviews or an existing AI platform) for comparative effectiveness evaluation. Such a robust design is essential to validate the pedagogical efficacy and scalability of the IIAA system across varied user demographics and educational contexts.

Furthermore, a vital research direction involves validating the LLM's automated assessment. This will be achieved by incorporating a human expert panel (e.g., HR professionals) to independently rate interview performance and responses. Future studies will then report inter-rater reliability metrics (such as Cohen's Kappa) to objectively validate the LLM's automated scoring and feedback quality against human judgment, thereby enhancing the trustworthiness of the system's evaluation component.

Finally, the integration of more sophisticated user feedback mechanisms and adaptive learning algorithms could substantially augment the pedagogical effectiveness and personalization of the training tool. Finally, extending linguistic support to other languages and incorporating a wider range of job roles and interview scenarios would significantly broaden the system's applicability and potential impact.

VI. CONCLUSION

This research detailed the development and initial evaluation of an advanced, game-based simulation tool designed

for professional job interview preparation. By integrating sophisticated technologies, notably MetaHumanSDK for high-fidelity TTS and STA functionalities, the system successfully generated interactive and perceptually realistic interview scenarios. The findings indicated that this approach enhanced user engagement and provided valuable, AI-driven feedback to participants.

While the study identified certain performance limitations, including response latencies associated with avatar rendering and challenges in speech detection for mixed Thai-English linguistic inputs, the innovative integration of these technologies within a serious game framework represents a notable advancement. This work contributes to the field of game-enhanced learning by demonstrating a viable methodology for creating immersive and effective professional skills training environments. Continued refinement of the system and expansion of its capabilities are anticipated to further advance its utility in professional skills development.

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CONFLICTS OF INTEREST

None

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